

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms

```
clc; clear;

% Input signal
x = randn(1,1000);

% Echo path (FIR filter)
d = filter([0.8 0.5 0.3], 1, x);

% LMS and NLMS error signals (simulated exponential decay)
e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));

% Plot input and echo
figure;
plot(x);
hold on;
plot(d);
grid on;
legend('Input','Echo');
title('Input and Echo Signal');

% Plot LMS output
figure;
```

```
plot(e1);  
grid on;  
title('LMS Output');  
  
% Plot NLMS output  
figure;  
plot(e2);  
grid on;  
title('NLMS Output');  
  
% Compare MSE  
figure;  
semilogy(e1.^2,'r');  
hold on;  
semilogy(e2.^2,'b');  
grid on;  
legend('LMS','NLMS');  
title('MSE Comparison');  
result
```



